

Preliminary Report

Introduction

This project is a privacy-focused tool designed to help individuals develop deeper self-awareness of their emotions and mental states. Users engage with an emotional stimulus, a short video of carefully chosen images, while their facial expressions are analysed in real time. Their verbal responses are then transcribed and analysed for emotional tone. The system synthesises this data to generate a personalised psychological profile, identify emotional patterns, and provide targeted recommendations for stress management, self-improvement, and mental wellness. All processing occurs locally on the user's device with zero data transmission. This is an introductory tool for people who lack access to therapy or prefer an impersonal first step. It is not a replacement for professional mental health care.

The project is based on Template 4.1: Orchestrating AI models to achieve a goal.

Most people struggle to understand their own emotions. Professional support is expensive and not always accessible. Mental health disorders constitute five of the top ten causes of disability across Europe, with an estimated total cost exceeding €600 billion annually (Laidi et al., 2023). This tool targets two groups: those who cannot afford professional therapy, and those who experience anxiety or stigma when discussing mental health with strangers.

Similar technology already exists but serves different purposes. Affectiva, a commercial product born out of MIT Media Lab, uses computer vision to detect emotions from facial expressions in real time. However it is built for corporate use, its primary application is measuring consumer reactions to advertisements. The insights go to the advertiser, not the person being analysed. It is expensive, closed source, and inaccessible to individuals.

A more recent example is a 2025 academic paper from Brunel University London, which combined facial emotion recognition, speech-to-text, and sentiment analysis, feeding everything into a language model, a pipeline technically very similar to this project. However their system only processes pre-recorded video, is not interactive, and uses GPT-4o, a paid cloud service. The authors themselves flagged this as a privacy concern.

This project addresses both gaps: it uses a live webcam feed instead of pre-recorded video, and runs the language model entirely locally using Ollama. Rather than producing a static report for a researcher or feeding data to an advertiser, the system opens an interactive conversation with the user, putting the insight back in the hands of the person it came from.

Literature review

Examples of similar projects

Abby AI, Medical Decision Support System

Abby AI is an Australian medical tool that uses AI to prepare patient profiles before clinical appointments. It collects medical history, past test results, and medication records, synthesises them, and presents a structured summary to the doctor before they see the patient (Abby Health, 2026). The doctor remains in control at all times, Abby does not diagnose or prescribe.

Both systems collect information about a person and generate a profile from it. The difference is who that profile is for. Abby serves the clinician, the insight goes to the professional, not the patient. This project does the opposite: there is no doctor, no intermediary. The profile goes directly to the user.

Affectiva, Emotion AI Software

Miller (2017) describes Affectiva as a company that makes AI software to detect human emotions, it started from MIT Media Lab. The company analyzes facial expressions, voice tone, and body movements to understand what people feel. And according to an article from Affectiva themselves (Affectiva, 2016) they used the Facial Action Coding System (FACS), developed by Paul Ekman, to label facial expressions.

Affectiva uses cameras and microphones to detect emotions in real-time. It analyzes facial expressions, voice tone, and body language. The company has studied over 10 million face videos from 90 countries. It can detect emotions like happiness, anger, confusion, and frustration.

My project aims at using similar technology, also based on Ekman work, to detect emotions in real time from a live feed, and give immediate feedback to the user, for free.

Affectiva is used by large companies for different purposes. Advertising companies use it to measure how people react to ads. Car companies use it to monitor drivers and detect if they are tired or distracted. Companies use it to understand customer emotions in call centers. Gaming and education companies use it to detect if users are bored.

Both Affectiva and my project use AI to detect emotions. But it is a paid software for big corporations to make more money and understand customers. My project is a free software for people to understand their own emotions better.

Techniques and methods

For my project I will combine several pre-trained models, to achieve a common goal. The domains I want to cover are four, vision, audio, text and language generation. Next is a list of the library and models I will initially explore. I will first try more demanding models that require higher resources and later decide whether it is appropriate or downgrade to more basic frameworks.

OpenCV, Computer Vision Library

OpenCV is an open-source library for detecting faces and tracking movement in videos. It runs locally on the user's machine, so the data stays private.

The library is cross-platform and licensed as free and open-source software (OpenCV, 2026) making it perfect for my project.

I will also explore other options, like MediaPipe. Some users say it's faster and simpler for facial detection (Harsha, 2026).

Whisper, Speech-to-Text Model

For speech recognition I will first try Whisper which is an open-source speech recognition model created by OpenAI (Radford et al., 2022). It's trained on 680,000 hours of multilingual audio data and works with 99 different languages. It runs locally on the user's machine without needing internet connection, perfectly aligning with my project standards.

Whisper will transcribe what the user says during the emotional assessment. The transcribed text will then be analyzed for sentiment and emotional tone (Radford et al., 2022).

I will also explore other options, like Vosk, which is an offline speech recognition framework that is lighter and faster than Whisper (Vosk, 2024).

Sentiment Analysis, Text Emotion Detection

To detect emotions and opinions in text I will feed the transcribed speech from Whisper to identify emotional tone and sentiment in the user's words using a pretrain model from Hugging Face.

The models can be downloaded and run locally on the user's machine. These models use deep learning to classify text as positive, negative, or neutral, and can detect more nuanced emotions like frustration, joy, or sadness (Hugging Face, 2024).

I will also explore VADER (Valence Aware Dictionary and Sentiment Reasoner), which is a simpler, rule-based approach that is lighter and faster than deep learning models (Hutto & Gilbert, 2014).

Ollama, Local Language Model Framework

Finally for the interactive part I will use Ollama, a framework that allows large language models to run locally on the user's computer without needing internet connection or cloud services (Ollama, 2024).

For my project, Ollama will generate the final psychological profile and personalized recommendations. The ideas will be based on all the data collected: facial emotions, transcribed speech, and sentiment analysis. Since Ollama runs entirely on the user's machine, all emotional data stays private and is never sent to external servers (Ollama, 2024).

Ollama supports different model sizes, from small models that run on basic computers to larger, more advanced models. I will also explore LM Studio, which is another local framework that is user-friendly and allows easy switching between different language models (LM Studio, 2024).

Study Cases and Research Evidence

To assess the effectiveness of AI in Psychological Diagnosis I relied on article from Alyssa Ezell who conducted a research study at Widener University testing whether six AI chatbots, three general-use (ChatGPT, Gemini, Claude) and three therapist-specific (Pi, Panda, Abby), could accurately diagnose mental health conditions and create treatment plans (Ezell, 2026).

General-use chatbots provided diagnoses in 100% of cases, but only 36.7% were completely correct; in 40% of cases, chatbots gave both correct and incorrect diagnoses simultaneously (Ezell, 2026). Therapist-specific chatbots were more cautious, providing diagnoses only 53.3% of the time (Ezell, 2026). Treatment planning, general-use chatbots recommended treatment plans in 93.3% of cases, most commonly suggesting Cognitive Behavioral Therapy, psychiatric medication, and social support, but these recommendations were very generic (Ezell, 2026).

Ezell's research demonstrates that AI can indeed provide mental health information but current systems struggle with accuracy, personalization, and transparency.

My project aims at bridging those gaps through real-time emotion detection, individual focus, and privacy-preserving local processing.

Design

Domain

Laidi et al. (2023) report that mental disorders constitute five of the top ten causes of disability across Europe, with an estimated total cost exceeding 4% of GDP, more than €600 billion annually. They also note that these conditions often begin early in life, with significant impacts on work productivity and quality of life.

The primary users of my product are individuals aged 18-45 seeking mental health support in Europe. The scale of the problem is considerable: in France alone, Laidi et al. (2023) estimate the annual cost of mental health at €163 billion in 2018, broken down into direct healthcare costs (14%), productivity loss (27%), and loss of quality of life (51%). These costs reflect both the prevalence of mental health conditions and barriers

to access, with anxiety and depressive disorders being the most common conditions, affecting millions of people across the continent.

Affordability is not the only issue though, discomfort with human interaction is also a critical barrier. While people generally prefer human doctors to AI systems, recent evidence shows that interaction with a fully automated AI chatbot can be an effective approach in the diagnosis and treatment of psychiatric disorders (Lee et al., 2022), some individuals may engage with AI-based tools when human interaction feels inaccessible or uncomfortable (Riedl et al., 2024).

This tool targets two user groups: those who cannot afford professional therapy, and those who experience anxiety or stigma when discussing mental health with strangers. The system is designed to assess the severity of a case, if a user's profile indicates serious mental health, the system actively encourages them to seek professional help. It is not a replacement for therapy, but a meaningful first step toward self-awareness and care-seeking.

Design Justification

The design choices in this project are based on the specific needs identified in the domain and user analysis: privacy, affordability, accessibility, and responsible care-seeking.

Fully local processing

To address privacy concerns that are especially important in mental health contexts, the entire system runs locally, with no internet connection or cloud services. Riedl et al. (2024) found that perceived privacy invasion was higher in AI-based interactions, and higher still in psychiatry than in other medical fields, where patients fear data breaches when discussing personal conditions. By keeping all emotional data on the user's machine, the system removes this barrier entirely. This is also what separates the project from commercial products like Affectiva, which send insights to third parties rather than the user.

Multiple models across different domains

The system combines facial emotion recognition, speech transcription, sentiment analysis, and a local language model. This is because people often find it hard to put their feelings into words, and what they say does not always match what their face shows. Using more than one source of information gives a fuller picture than any single model could. Recent work in the field follows the same logic: the Brunel University study (2025), for example, combined facial emotion recognition with text sentiment because no single model captured the full emotional picture on its own.

Interactive conversation

After analyzing users' emotions the system opens a dialogue through the local language model instead of producing a static diagnosis. This matters because the target users are often approaching mental health for the first time and need guidance, not just a label. An interactive format lets them ask questions, explore recommendations, and feel heard. Lee et al. (2022) found that AI chatbots can foster rapport and self-disclosure, which is valuable for users who feel anxious or embarrassed talking to a stranger.

Severity assessment

It is important to notice that the system is not designed to replace therapy. When a user's profile suggests serious concerns, it encourages them to seek professional help. The evidence is clear, people prefer and benefit from human care (Riedl et al., 2024), and an AI tool that masks the need for real treatment would be harmful.

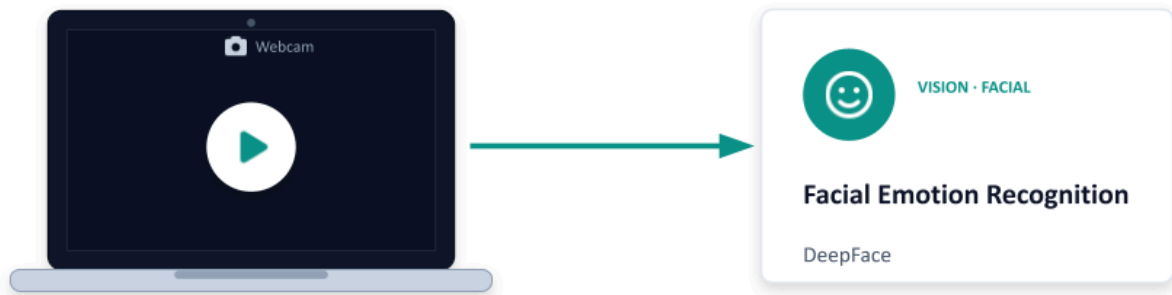
This project recognises the importance of human connection in effective mental healthcare, and it must be considered as a first step rather than a substitution.

Structure of the project

Emotional AI mirror

1 Watch the Video

The webcam reads the user's facial expressions in real time.

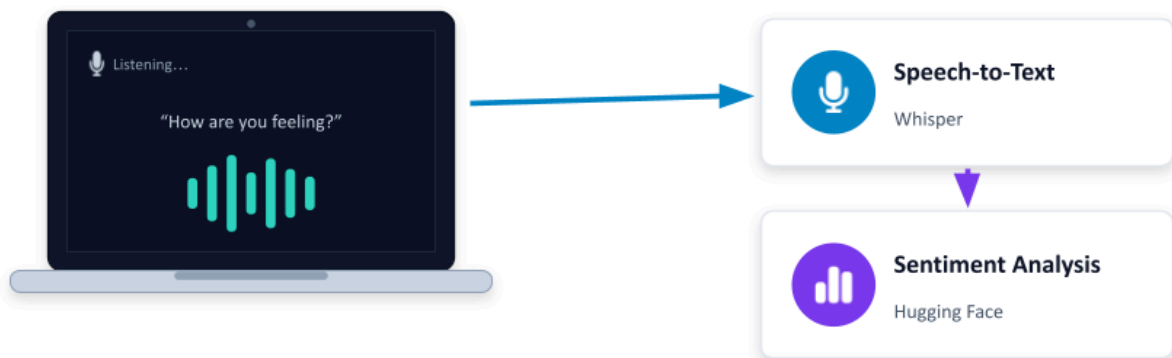


The user sits in front of their laptop and watches a short sequence of emotionally provocative images. While this happens, the webcam captures live video with the user's consent and feeds it to a facial emotion recognition model (DeepFace), which detects how the user's face reacts in real time and classifies expressions into emotional states.

Emotional AI mirror

2 Answer Out Loud

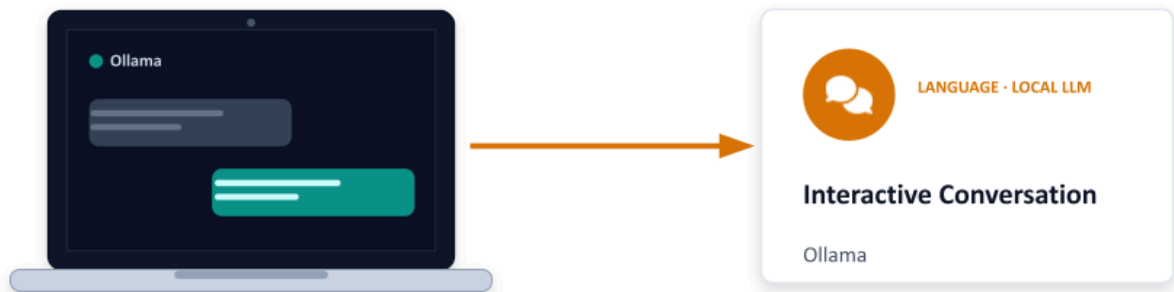
The user answers a few questions aloud, the voice is analysed.



The system then asks the user a few questions, which they answer aloud. Their voice is transcribed by Whisper, OpenAI's speech-to-text model running locally. The resulting text is passed to a Hugging Face sentiment analysis model, which identifies the emotional tone behind what the user said.

3 Chat

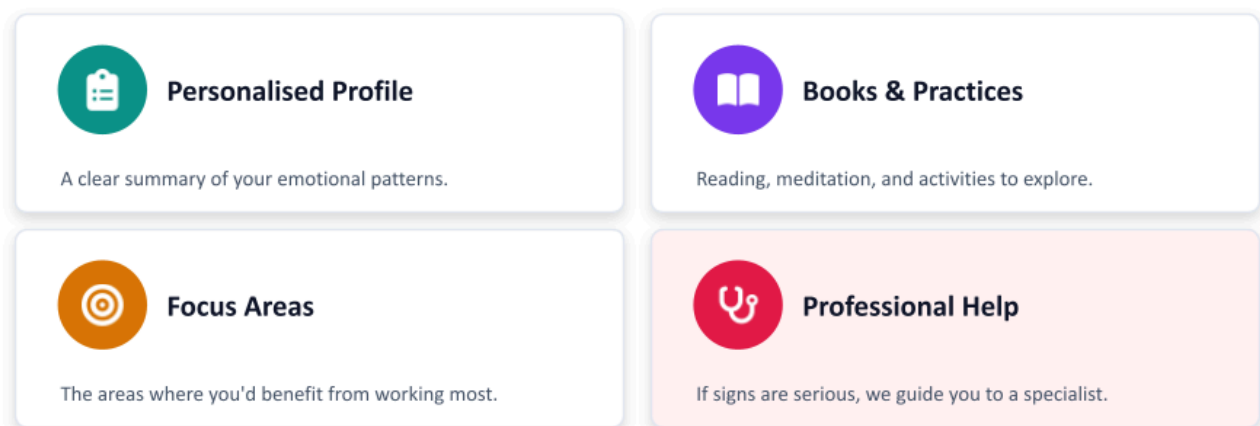
A local AI discusses what it observed and offers guidance.



All the data collected so far: facial reactions, transcribed answers, sentiment scores; is then fed to a local language model running through Ollama. The model synthesises everything it has observed and opens an interactive conversation with the user, letting them ask questions and explore their results.

4 Your Personalised Profile

Personal insights, resources, and clear next steps.



Finally the system delivers a personalized psychological profile which consists of a summary of detected emotional patterns, reported concerns, and personalised wellbeing recommendations, alongside practical recommendations: books to read, meditation practices, focus areas to work on, and lifestyle suggestions. If the profile indicates serious concerns, the system actively encourages the user to seek professional help from a qualified therapist.

Technologies and methods

The project will be built in Python, chosen for its vast ecosystem of machine learning libraries and support for the pre-trained models involved. Four AI models form the core of the system: a facial emotion recognition model (DeepFace) for analysing webcam footage, Whisper for local speech-to-text transcription, a Hugging Face transformer model for sentiment analysis, and a local language model through Ollama for profile generation and conversation. Supporting libraries include OpenCV for capturing and processing the live webcam feed, and the Hugging Face Transformers library for running the sentiment model. The key methods stitching these together are real time video processing for facial analysis, and combining the visual, audio, and text signals so the language model can interpret them together as a single picture of the user. All processing runs locally to preserve privacy.

A plan of work

Project Plan	July to September 10, 2026										
Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11
	Jun 29	Jul 6	Jul 13	Jul 20	Jul 27	Aug 3	Aug 10	Aug 17	Aug 24	Aug 31	Sep 7
Project setup and environment											
Test and select facial emotion model											
Test and select speech-to-text model											
Test and select sentiment analysis model											
Test and select local LLM (Ollama)											
Integrate the four models into a pipeline											
Build user interface and interaction flow											
Prompt engineering and profile generation											
User testing and iteration											
Final report and submission											
MILESTONES						All models selected		Working prototype		Submission	

Test and evaluate

Since this project uses pre-trained models instead of building one from scratch, evaluation is less about large user studies and more about whether the models run well on my hardware and work together as one pipeline. Testing is split into three stages: checking each model on its own, testing how they connect, and getting feedback from real users.

The first stage checks each model in isolation. For every model I will read the documentation for the minimum hardware required, install it locally, and run it on sample inputs to confirm the output is usable. I will record the processing time per input and whether the model runs within available memory, so the decision is based on real numbers. If a model is too slow or too heavy for my hardware, I will swap it for a lighter one, for example Vosk instead of Whisper, or VADER instead of a transformer-based sentiment model. This swap is the main fallback built into the design.

The second stage tests how the models connect. The main concern here is that the output of one model reaches the next in a usable form: that Whisper's transcription passes cleanly to the sentiment model, and that all the signals arrive at the language model correctly. I will look for specific problems like mismatched data formats, empty or dropped inputs, and the total processing time across the whole pipeline, to make sure the system works from start to finish.

The third stage tests the experience with real users. Between six and ten testers, picked among friends and relatives, will go through the full process and then rate the result with a short Likert-scale survey (1-5) on

how accurate, clear, and useful they found the profile and recommendations. Each tester also fills in a short self-report beforehand, which I compare against the profile the system generates to see how well the two match. This gives me a measurable signal of quality instead of relying only on impressions, and I will use the feedback to improve the questions, the video, and the language-model prompts.

The project will be judged against the goal I set at the start: a system that runs entirely locally and produces a coherent, useful psychological profile. In practice, this means the full four-model pipeline runs reliably at acceptable speed on a normal computer, the average Likert ratings for usefulness and clarity are high, and the profile matches the user's self-report reasonably well. Falling short on any of these points to where the work is needed: low agreement suggests the prompts or model weighting need fixing, while low usability ratings point to the interaction flow.

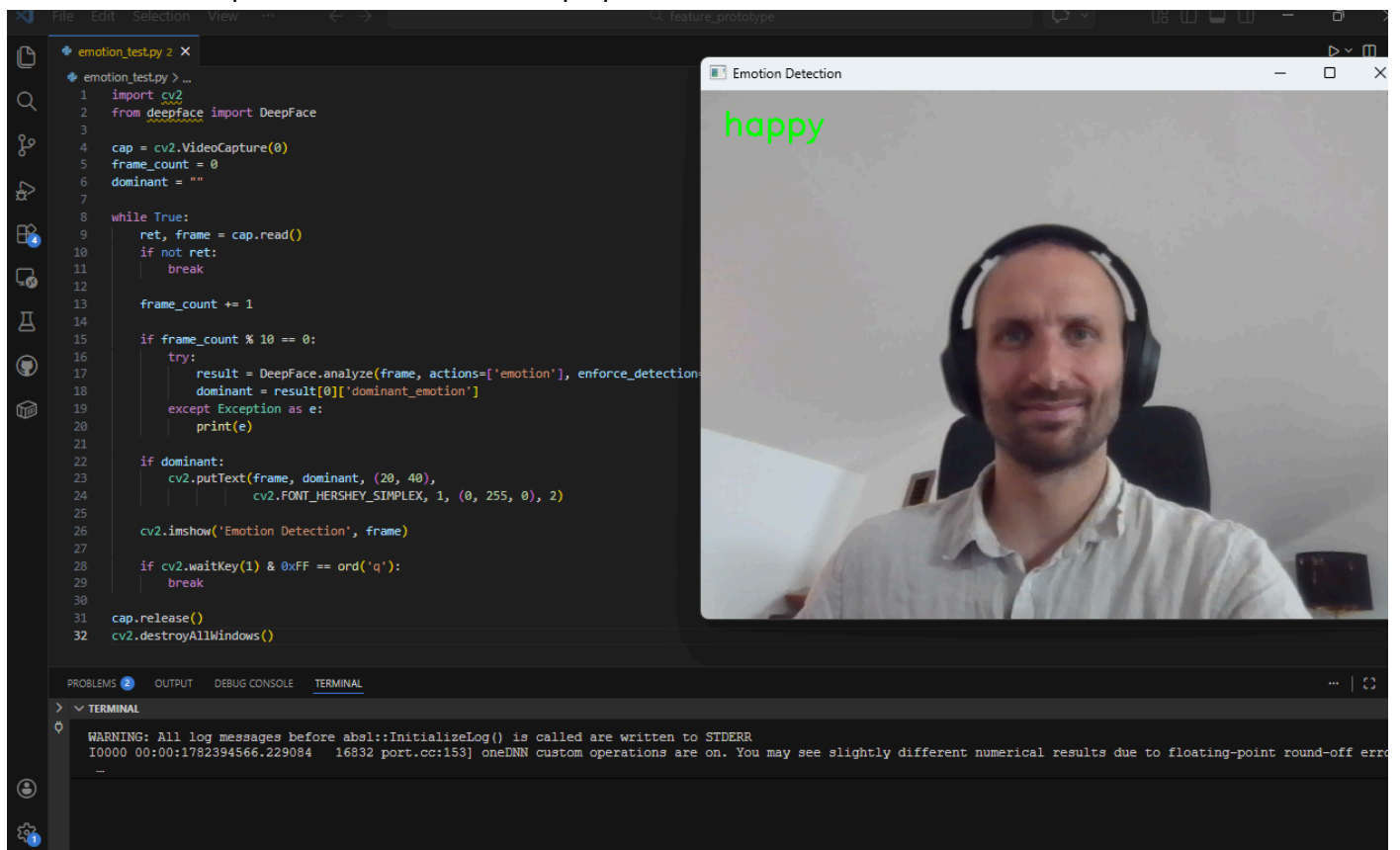
Feature Prototype

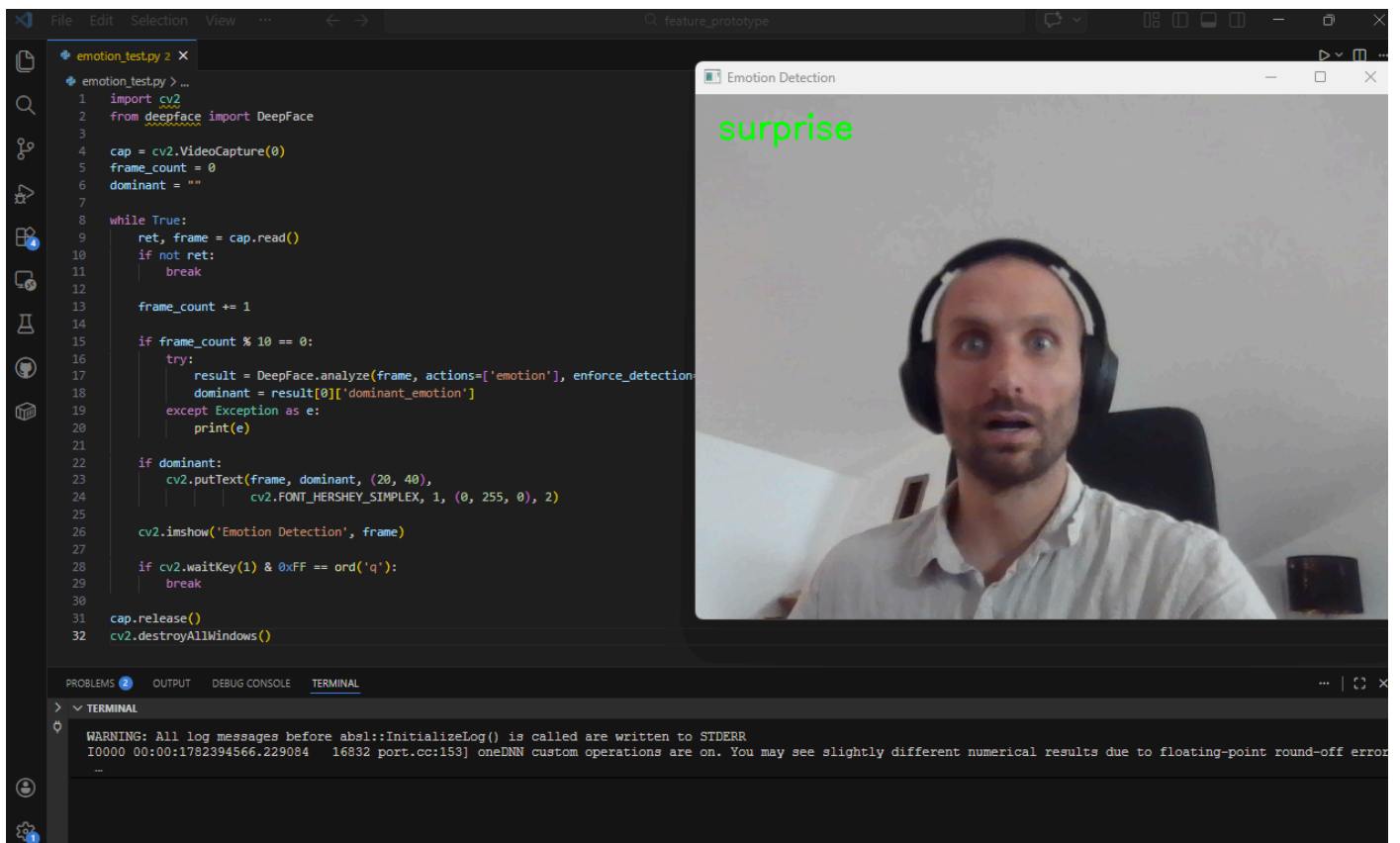
Facial emotion recognition pipeline. Live webcam → emotion detection → emotion timeline.

Implementation

DeepFace

The first step was to verify that DeepFace could run on the available hardware. A minimal script was written using DeepFace and OpenCV to capture live webcam footage and display the detected dominant emotion on screen. The goal at this stage was not accuracy or performance, but simply to confirm the model was functional and responsive on a consumer laptop with no dedicated GPU.

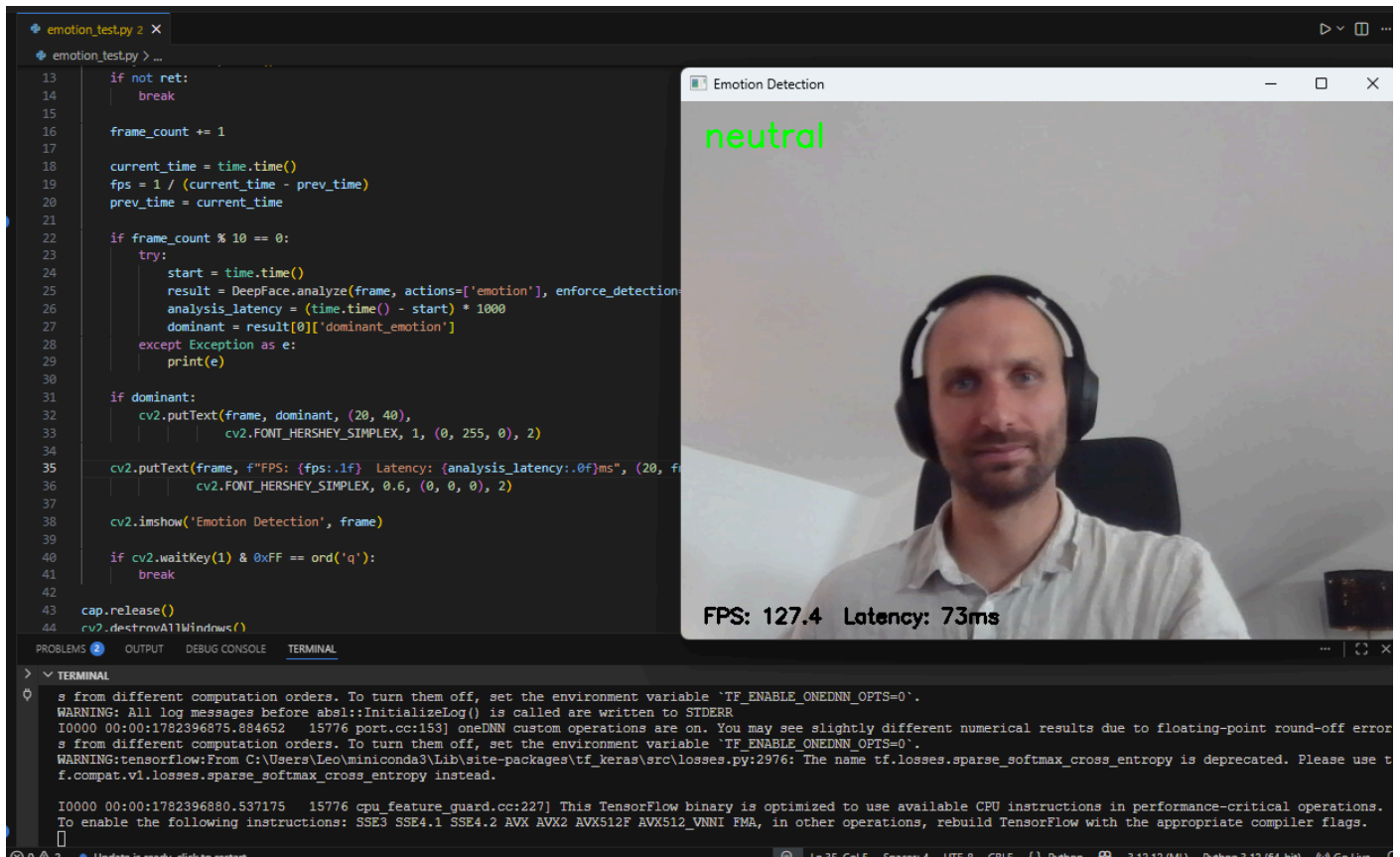




The model loaded and ran successfully on the test machine, an Intel Core i7-1185G7 with 16GB RAM and no dedicated GPU, confirming that local execution is viable without specialist hardware.

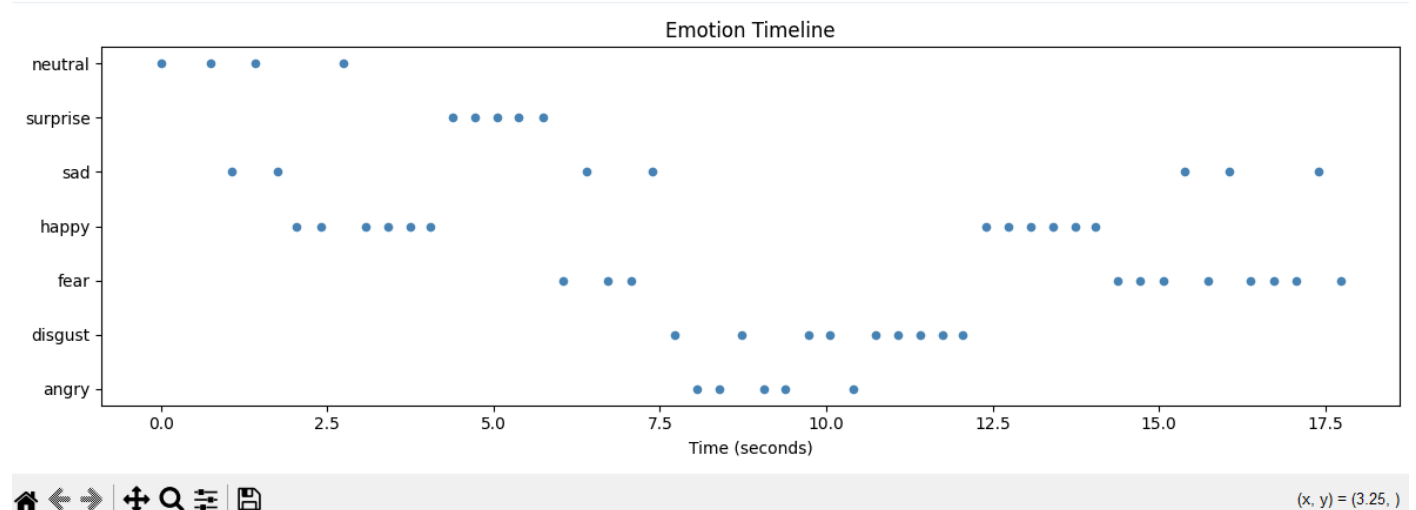
FPS

To measure performance, FPS and analysis latency were added to the script. Running on an Intel Core i7-1185G7 with 16GB RAM and no dedicated GPU, the system achieved approximately 127 FPS for video capture and 73ms per emotion analysis. Since analysis only runs every 10 frames, the display remains fluid while the CPU load stays manageable. These results confirm the pipeline is fast enough for real-time use on consumer hardware.



Emotion Timeline

The graph shows the model picking up emotion changes in real time. When a clear expression was made, smiling, raising eyebrows, frowning, the model detected it correctly. Between expressions, when the face was relatively neutral, the model often returned fear or disgust instead of neutral. This suggests the model struggles with resting or ambiguous faces, and tends to over-detect negative emotions when there is no strong signal.



Linking emotions

For the purposes of this prototype, the video stimulus is simulated using a predefined sequence of labelled scenes with fixed timestamps. The content of the scenes is not psychologically calibrated at this stage. The goal is solely to verify that emotion readings can be correctly tagged to a scene label and passed to the language model in a structured way. Selecting and sequencing meaningful stimulus content is left for a later stage of development.

```
# Simulated video sequence (start, end, label)
video_sequence = [
    (0, 5, "threatening scene"),
    (5, 10, "joyful scene"),
    (10, 15, "sad scene"),
    (15, 20, "relaxing scene"),
]

def get_current_scene(elapsed):
    for start, end, label in video_sequence:
        if start <= elapsed < end:
            return label
    return "end"
```

The `video_sequence` list defines the simulated stimulus timeline. The `get_current_scene` function checks the elapsed time and returns whichever scene label is currently active.

The script ran for 20 seconds, cycling through four labelled scenes. For each scene the user deliberately expressed a matching emotion. The model correctly identified the dominant emotion in all four cases: surprise during the threatening scene, happy during the joyful scene, sad during the sad scene, and neutral during the relaxing scene. This confirms that the emotion-to-scene tagging works as intended and produces structured data that can be passed to a language model.

Ollama

To enable local language model inference, Ollama was installed on the test machine. Ollama is a framework that allows large language models to run entirely locally without an internet connection (Ollama, 2024). The smallest available model, `llama3.2`, was pulled via the terminal using `ollama pull llama3.2`. The model is approximately 2GB and runs on CPU, requiring no dedicated GPU.

```
(base) 0 final project\feature_prototype> cmd.exe activate feature_prot
(feature_prot) 0 final project\feature_prototype> ollama pull llama3.2
pulling manifest
pulling d8c3a31c7ff: 100% 2.0 GB
pulling 966de95ca8a6: 100% 1.4 KB
pulling f0c3a9bec8da: 100% 7.7 KB
pulling 87d4f7e37089: 100% 6.0 KB
pulling 58bb8bd477a5: 100% 96 B
pulling 34bb5ab01051: 100% 561 B
verifying sha256 digest
writing manifest
success
(feature_prot) 0 final project\feature_prototype> █
```

Following the webcam session, the emotion summary was passed to `llama3.2` running locally through Ollama. The model received a structured prompt describing the dominant emotion detected during each scene and returned an initial psychological observation. The response demonstrated that the language model can interpret emotion data in a meaningful way and generate contextually relevant commentary. The quality of the output is limited at this stage by the simplicity of the prompt and the placeholder nature of the stimulus content, both of which will be refined in the final system.

Evaluation

The prototype successfully demonstrates the core pipeline: facial emotion detection runs in real time on consumer hardware, emotion readings are correctly tagged to scene labels, and the resulting summary is interpreted by a local language model to produce an initial psychological observation. FPS averaged around 127 with an analysis latency of approximately 73ms per frame, confirming the system is responsive enough for real-time use.

Emotion detection was accurate when expressions were deliberate and clear. However the model showed a tendency to default to negative emotions such as fear and sadness during neutral or ambiguous expressions, which is a known limitation of frame-by-frame analysis with no temporal context.

Limitations and Future Work

The stimulus content is currently simulated with placeholder labels rather than real psychologically calibrated images or video. The language model prompt is basic and would need significant refinement to produce meaningful output. The system also lacks the speech transcription and sentiment analysis components, which in the final version will provide additional context to the language model alongside the facial data. Finally, the interaction is currently one-way, the language model responds once and the session ends. The final system will open a full interactive conversation with the user.